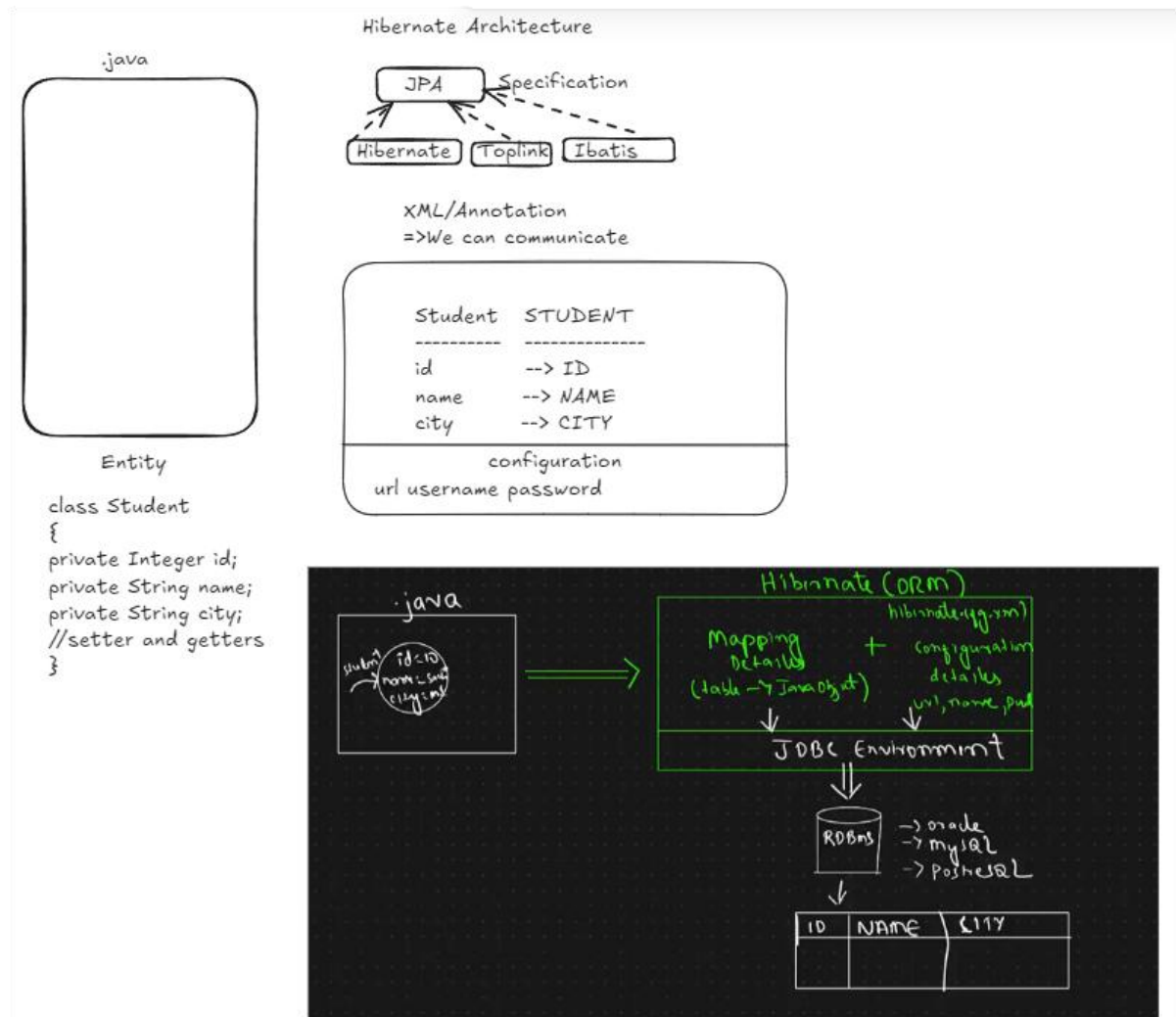


Hibernate Architecture

Hibernate is a powerful, high-performance Object-Relational Mapping (ORM) tool that simplifies the interaction between Java applications and relational databases. It abstracts the database operations by allowing developers to focus on the business logic rather than SQL queries.

Hibernate Architecture image



Core Components of Hibernate Architecture:

1. JPA (Java Persistence API) Specification:

- Hibernate implements the JPA specification, which standardizes ORM in Java.

- It provides a common interface for different ORM tools like Hibernate, Toplink, and iBatis.

2. Mapping Configuration:

- Hibernate uses XML files or annotations to map Java objects to database tables.
- Each Java class that needs to be persisted in the database is known as an **Entity**.

```
class Student {  
  
    private Integer id;  
  
    private String name;  
  
    private String city;  
  
    // Getters and setters  
  
}
```

The class Student is mapped to the STUDENT table in the database with fields id, name, and city corresponding to the columns ID, NAME, and CITY.

Hibernate Configuration:

- Hibernate configuration involves setting up the database connection details such as the URL, username, and password.
- This configuration can be done through an XML file (hibernate.cfg.xml) or programmatically via annotations.

Hibernate ORM (Object-Relational Mapping):

- Hibernate converts Java objects into database records (rows in tables) and vice versa.
- The process of converting a Java object into a database record is called **Mapping**. This involves:

- **Mapping Details:** Java objects are mapped to relational database tables.
- **Configuration Details:** These include the database connection properties and other settings needed for the ORM tool to function correctly.
- Hibernate interacts with the **JDBC Environment** to execute SQL queries and manage database transactions.

Database Connectivity:

- Hibernate can work with various relational databases (RDBMS) like Oracle, MySQL, PostgreSQL, etc.
- The ORM tool abstracts the underlying database operations, allowing developers to switch between different databases with minimal configuration changes.

Entity & Table Mapping:

- The entities (e.g., Student) in Hibernate are directly mapped to corresponding database tables (STUDENT).
- Each field in the entity class (e.g., id, name, city) is mapped to a column in the database table.

Process Flow in Hibernate:

1. Java Object Creation:

- A Java object (e.g., Student) is created with data.

2. ORM Handling:

- Hibernate ORM handles the conversion of this object into a database record by using the mapping configuration.

3. Database Operation:

- Hibernate interacts with the JDBC environment to insert, update, or query the database.

- The data is stored in the appropriate table (e.g., STUDENT table) in the chosen RDBMS.